ALA - 2 Python Deep Dive: Interview Intensive

Topic: Exception Handling in Python (try-except)

Interview Related Questions

1. What is try-except in Python?

Definition:
try-except is used to handle runtime errors and prevent the program from crashing.

Syntax:

```
try:
   # risky code
except:
   # handling code
```

Example:

```
try:
   print(10 / 0)
except:
   print("Error occurred")
```

2. What is try block?

Definition:
The try block contains code that may cause an error.

Syntax:

- **try:**
- **statement**

Example:

```
try:
    x = int("abc")
```

3. What is except block?

Definition:
The except block handles the error generated in the try block.

Syntax:

```
except ExceptionType:
    statement
```

Example: try:

```
    x = int("abc")
except ValueError:
```

```
    print("Invalid input")
```

4. What is multiple except in Python?

Definition:
Multiple except blocks are used to handle different types of errors separately.

Syntax:
```
try:
    statement
except Error1:
    statement
except Error2:
    statement
```

Example:
```
try:
    x = int(input())
    print(10 / x)
except ValueError:
    print("Invalid input")
except ZeroDivisionError:
    print("Cannot divide by zero")
```

5. What is else block in try-except?

Definition:
The else block executes only if no error occurs in the try block.

Syntax:
```
try:
    statement
except:
    statement
else:
    statement
```

Example:
```
try:
    x = int(input())
except:
    print("Error")
else:
    print("No error")
```

6. What is finally block?

Definition:
The finally block always executes whether an error occurs or not.

Syntax:
```
try:
    statement
```

```
except:
    statement
finally:
    statement
```

Example:

```
try:
    f = open("file.txt")
except:
    print("Error")
finally:
    print("Done")
```

7. What is Exception as e?

Definition:
It is used to capture the error message in a variable.

Syntax: except Exception as e:

Statement

Example:

```
try:
    print(10 / 0)
except Exception as e:
    print("Error:", e)
```

8. What is nested try-except?

Definition:

A try-except block inside another try block is called nested try-except.

Syntax:
```
try:
    try:
        statement
    except:
        statement
except:
    statement
```

Example:
```
try:
    try:
        print(10 / 0)
    except ZeroDivisionError:
        print("Inner error")
except:
    print("Outer error")
```

9. What is raise keyword?

Definition:
raise is used to generate an exception manually.

Syntax: raise ExceptionType("message")

Example:
```
try:
    age = -5
    if age < 0:
        raise ValueError("Invalid age")
except ValueError as e:
    print(e)
```

10. Write a program using try-except

Definition:
This involves handling errors using try-except blocks.

Example:
```
try:
    num = int(input("Enter number: "))
    print(10 / num)
except ZeroDivisionError:
    print("Cannot divide by zero")
```

```
except ValueError:
    print("Invalid input")
```